QP-25901052

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION November 2025

First SEMESTER

Skill Development Components (SDC) - MC1SDCREM101

- FUNDAMENTALS OF RENEWABLE ENERGY TECHNOLOGY (2025 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Rememder {Xj, *Understand(U), Apply(A), Analyse(An),* Ev'a/oate{E/, Czeate{C/, Sir///{S}, *Interest(I) and Appreciation(Ap)*

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PART A

Short answer

Answer any 15 questions

*Each question carries **1** marks*

1. **Recall one primary energy source. [K]/[CO1]**

   Sunlight is a primary energy source that provides energy naturally.

2. **Identify the cause of wind energy. [K]/[CO1]**

   Wind energy is caused by the movement of air due to uneven heating of the Earth's surface.

3. **State the main source of fossil fuels. [K]/[CO1]**

   Fossil fuels are mainly derived from dead plants and animals buried over millions of years.

4. **Define the greenhouse effect. [K]/[CO1]**

   The greenhouse effect is the trapping of heat in the Earth's atmosphere by greenhouse gases.

5. **Identify the physical principle involved in converting solar radiation into heat.[U]/[CO2]**

Solar radiation is absorbed by a surface and converted into heat through the principle of thermal energy conversion.

6. **Inspect the role of the reflective surface in a concentrator.[An]/[CO4]**

   The reflective surface in a concentrator focuses sunlight onto a receiver to increase heat intensity.

7. **Outline gases evolved during dry-process conversion. [An]/[CO4]**

   Dry-process conversion produces gases such as carbon dioxide, carbon monoxide, and methane.

8. **Inspect the importance of glazing transmittance on heat gain.[An]/[CO4]**

   Higher glazing transmittance allows more solar heat to enter, increasing the efficiency of heat gain.

9. **Implement the factor that increases drag force on a body moving through air.[A]/[CO3]**

   Drag force increases when the velocity of the body or its surface area exposed to air increases.

10. **Execute the main function of the rotor blades in a wind energy system.[A]/[CO3]**

    Rotor blades capture the wind's kinetic energy and convert it into rotational mechanical energy.

11. **Categorize hybrid systems that integrate wind and solar.[An]/[CO4]**

    Systems that combine wind turbines and solar panels to generate electricity are called wind-solar hybrid systems.

12. **Compare the roles of the rotor blades and the generator in a wind energy system.[An]/[CO4]**

    Rotor blades convert wind energy into mechanical energy, while the generator converts that mechanical energy into electricity.

13. **Differentiate between energy crops and municipal solid waste. .[An]/[CO4]**

    Energy crops are specially grown plants for fuel, whereas municipal solid waste is organic waste collected from households and cities.

14. **Compare batch and continuous digestion systems. .[An]/[CO4]**

    Batch digestion processes material in a single batch, while continuous digestion processes feedstock continuously for energy production.

15. **Examine the potential of algae-based bioenergy systems. .[An]/[CO4]**

    Algae-based bioenergy systems can produce biodiesel and bioethanol efficiently due to high growth and oil content.

16. **Examine how biomass energy supports rural and local economies. .[An]/[CO4]**

    Biomass energy creates local jobs and income opportunities, helping rural communities develop economically.

17. **Inspect methanogenic bacteria roles in digestion. .[An]/[CO4]**

    Methanogenic bacteria break down organic matter in anaerobic conditions to produce methane gas during digestion.

PART B

Short Essay

Answer any 3 questions

*Each question carries **5** marks*

18. **Recall the main characteristics of coal, petroleum, and natural gas as conventional energy sources. [K]/[CO1]**

    Coal is a solid fossil fuel with a high carbon content and is mainly used for electricity generation and industrial heating. Petroleum is a liquid fuel that is easily refined into products like gasoline, diesel, and kerosene, making it highly versatile for transportation and industry. Natural gas is primarily composed of methane and is considered a cleaner fossil fuel due to lower carbon dioxide emissions when burned. All three are non-renewable and take millions of years to form from ancient organic matter. They are energy-dense and provide reliable, continuous power. However, their extraction and use can cause environmental issues, including air pollution and greenhouse gas emissions. These fuels have played a major role in global industrialization and modern energy systems.

19. **Breakdown the construction of a flat-plate collector and inspect how each component contributes to heat absorption.[An]/[CO4]**

    A flat-plate collector typically consists of a transparent cover, an absorber plate, a fluid-carrying tube, insulation, and a casing. The transparent cover allows sunlight to pass through while reducing heat loss to the environment. The absorber plate, usually painted black, absorbs solar radiation and converts it into heat. The fluid-carrying tubes transfer the absorbed heat to a circulating fluid, which carries it to

storage or directly to use. Insulation at the back and sides prevents heat loss from the collector, improving efficiency. The casing protects the components from weather while supporting the structure. Each part works together to maximize heat absorption and minimize energy loss.

20. **Examine the site selection considerations for a windmill.[An]/[CO4]**

Site selection for a windmill requires an area with high and consistent wind speeds to ensure maximum energy generation. The terrain should be free from obstructions like tall buildings, trees, and hills that could reduce wind flow. Proximity to power lines and the grid is important for easy transmission of electricity. Environmental and social factors, such as wildlife impact, noise, and visual aesthetics, should also be considered. Accessibility for construction and maintenance equipment is essential. Local climatic conditions, including extreme weather events, must be assessed to ensure safety and durability. Proper site selection directly affects the efficiency, lifespan, and economic viability of a wind energy project.

21. **Compare thermochemical and biochemical conversion processes. .[An]/[CO4]**

Thermochemical conversion involves using heat to convert biomass into energy or fuels through processes such as combustion, gasification, and pyrolysis. Biochemical conversion uses microorganisms or enzymes to break down organic material, producing products like biogas, ethanol, or other biofuels. Thermochemical methods are generally faster and suitable for dry biomass, while biochemical processes are slower and more effective for wet organic matter. Thermochemical processes often require high temperatures, whereas biochemical processes operate under mild conditions. Thermochemical methods can produce a variety of fuels, including syngas, char, and bio-oil, while biochemical methods mainly produce methane and alcohols. Both methods help in energy recovery from waste and reduce reliance on fossil fuels. Choice of method depends on feedstock type, desired product, and economic feasibility.

22. **Categorize organic materials based on their biogas yield potential. .[An]/[CO4]**

Organic materials can be categorized into high, medium, and low biogas yield potential based on their composition. High-yield materials include food waste, animal manure, and energy crops, which are rich in carbohydrates and proteins. Medium-yield materials include crop residues and green leaves, which produce moderate amounts of biogas. Low-yield materials include lignin-rich materials like wood and sawdust, which are difficult to digest and produce minimal biogas. The biogas yield

depends on factors such as moisture content, carbon-to-nitrogen ratio, and biodegradability. Proper categorization helps in designing efficient biogas systems and improving gas production. Mixing materials from different categories can also enhance overall yield and maintain a balanced digestion process.

PART C

Essay

Answer any 2 questions

*Each question carries **10** marks*

23. **List and show the advantages of non-conventional energy sources for sustainable energy development. [K][CO1]**

Non-conventional energy sources include solar, wind, biomass, geothermal, small hydro, and tidal energy. These sources are renewable and naturally replenished, unlike fossil fuels which are finite. They produce little or no greenhouse gas emissions, helping to combat climate change and reduce air pollution. Solar energy can be used for electricity, heating, lighting, and water purification, especially in remote areas. Wind energy provides clean power for residential, industrial, and agricultural applications, depending on the location. Biomass energy utilizes agricultural residues, municipal waste, and animal dung, converting them into electricity, biogas, or biofuels, which also helps manage waste. Geothermal and tidal energy provide stable, continuous power in suitable regions. Non-conventional sources reduce reliance on imported fuels, enhancing energy security. They create employment opportunities in installation, operation, and maintenance, supporting local economies. Furthermore, these sources encourage sustainable development by balancing environmental, social, and economic needs. With government incentives and technological advancements, their efficiency and affordability are improving. Overall, non-conventional energy is essential for a sustainable, low-carbon energy future.

24. **Compare the types of solar photovoltaic systems.[An]/[CO4]**

Solar photovoltaic (PV) systems convert sunlight directly into electricity using

semiconductor materials. PV systems are broadly classified into three types: on-grid (grid-connected), off-grid (standalone), and hybrid systems.

On-grid PV systems are connected to the electricity grid. They supply electricity directly to the grid and can feed surplus energy back, often earning credits or revenue under net metering schemes. These systems do not require batteries for storage because the grid acts as a virtual storage. They are suitable for urban areas and industries where grid access is available. On-grid systems are cost-effective for large-scale power generation and reduce reliance on conventional electricity.

Off-grid PV systems, also called standalone systems, operate independently of the grid. They require batteries to store electricity for use during nighttime or cloudy periods. These systems are ideal for remote or rural areas without grid access. They are used to power homes, schools, irrigation pumps, and small community projects. Proper sizing of solar panels and batteries is essential to meet energy demand reliably.

Hybrid PV systems combine solar PV with other energy sources such as wind turbines, diesel generators, or battery storage. They provide continuous power supply even when solar energy is insufficient. Hybrid systems are suitable for areas with variable sunlight or critical loads like hospitals and communication towers. They offer flexibility, reliability, and higher efficiency but are more complex and expensive.

PV modules can be monocrystalline, polycrystalline, or thin-film, affecting efficiency, cost, and space requirements. The choice of system depends on factors like location, energy demand, budget, grid availability, and reliability requirements. All PV systems contribute to clean, renewable energy generation and reduce carbon footprints.

In conclusion, on-grid systems are efficient and economical for connected areas, off-grid systems ensure energy access in remote locations, and hybrid systems provide reliable and flexible power supply. The development of these PV systems plays a vital role in sustainable energy generation, rural electrification, and reducing dependence on fossil fuels.

25. **Outline various applications of wind energy in industrial and rural contexts. [An]/[CO4]**

Wind energy is a renewable and sustainable source of power suitable for both industrial and rural applications. In industrial contexts, large wind turbines generate electricity for factories, warehouses, and commercial operations, reducing dependence on conventional power. Wind energy can also supplement grid electricity, especially

in regions with variable demand. In rural areas, wind energy is widely used for water pumping, irrigation, and livestock watering, helping farmers maintain sustainable agriculture. Small wind turbines provide electricity for homes, schools, and community centers in remote villages. Hybrid systems combining wind and solar PV ensure continuous power supply in off-grid areas. Wind energy supports decentralization, reducing transmission losses and improving energy reliability. It creates local jobs in installation, maintenance, and manufacturing of wind turbines. Environmentally, it reduces greenhouse gas emissions and helps conserve fossil fuels. Additionally, rural communities become more energy self-reliant, promoting economic development. Wind energy can also be harnessed for desalination, milling, and cold storage applications, expanding its usefulness. The low operating cost after initial investment makes it economically viable in the long run.

26. **Examine how biomass can be converted into different forms of energy (solid, liquid, and gas) and outline the advantages of each method. .[An]/[CO4]**

Biomass can be converted into solid, liquid, and gaseous energy forms through various methods. Solid biomass includes wood, pellets, and briquettes, which are directly combusted to generate heat or electricity. This method is simple, cost-effective, and suitable for cooking, heating, and small-scale power generation. Liquid biomass energy, such as biodiesel and bioethanol, is produced through transesterification or fermentation. Liquid biofuels are easier to store, transport, and use in vehicles and machinery, serving as a renewable alternative to petroleum fuels. Gaseous biomass energy, produced via anaerobic digestion or gasification, yields biogas or syngas. Biogas can be used for cooking, electricity, and heating, while syngas can power engines and turbines. Each form of biomass energy helps reduce greenhouse gas emissions and dependence on fossil fuels. Solid biomass is reliable and locally available, liquid fuels provide high energy density, and gases are clean and versatile. Biomass energy also promotes rural development by creating employment in collection, processing, and distribution. By converting waste and residues into energy, it supports a circular economy. Advanced technologies allow co-generation of heat and power, improving overall energy efficiency. Using biomass reduces environmental pollution and contributes to sustainable energy systems.